Rajinder Lal
Math 400

Chapter 3 Non-Online Homework

Section 3.1 page 180 # 7, 14, 27, 29, 36, 56, 57, 71, 72, 81

Section 3.2 page 188 # 6, 18, 23, 24, 29, 44

Section 3.3 page 196 # 2, 3, 13, 30, 34, 40, 43, 44, 46, 47

Section 3.4 page 204 # 13, 26, 30, 33, 34, 36, 44, 47, 54, 67, 71, 84

Please include the following problems as part of your Section 3.4 Homework:

A. Find the values of x, where the tangent line to the graph of *f* is horizontal.

$f(x) = (7x - 2)^5(x + 6)^4$

B. Find the derivative of the following function.

$f(x) = \cos(x^3 + x^2 \tan(e^x))$

Section 3.5 page 215 # 14, 19, 37, 53, 59, 71

Please include the following problems as part of your Section 3.5 Homework:

A. Find $\frac{dy}{dx}$. (Hint: Use implicit differentitation.)

(a.) $\cos(x - y) + \cos(x + y) = xy$

(b.) $x^3y^2 - 10x^2 = 4x^2 + 3y^5$

Section 3.6 page 223 # 18, 22, 25, 30, 34, 42, 45, 48

Please include the following problems as part of your Section 3.6 Homework:

Instructions: Find the derivative of the following functions.

A. $y = x^{\tan\left(4x^2+1\right)}$

B. $y = \log_4 \sqrt{\dfrac{10-z^2}{10+z^2}}$

C. $y = \log_{10}\left(\log_3\left(2x+1\right)\right)$

D. $y = 3^x \log_3\left(x^2 \sin x\right)$

Section 3.7 page 235 # 29, 31, 35

Please include the following problem as part of your Section 3.7 Homework:

A. The position of a particle is given by the equation $s(t) = t^4 - 8t^3 + 18t^2$ where *t* is measured in seconds and *s* in meters.

(a.) Find the position at $t = 1$, $t = 2$, $t = 3$, and $t = 6$

(b.) Find the velocity at time t.

(d.) When is the particle at rest?

(e.) When is the particle moving in the positive direction? Negative direction?

(f.) Find the total distance traveled on the intervals $[0,1]$, $[0,2]$, $[0,3]$, and $[0,6]$.

(g.) Draw a diagram to represent the motion of the particle. (Similar to Figure 2 on page 225)

(h.) Find the acceleration at time t.

(i.) Graph the position, velocity, and acceleration functions.

(j.) When is the particle speeding up? When is it slowing down?

Solution to A (except (g.)) on the next few pages.

Solution to Section 3.7 A

A. The position of a particle is given by the equation $s(t) = t^4 - 8t^3 + 18t^2$ where *t* is measured in seconds and *s* in meters.

(a.) Find the position at $t = 1$, $t = 2$, $t = 3$, and $t = 6$

$s(1) = (1)^4 - 8(1)^3 + 18(1)^2 = 11$ meters

$s(2) = (2)^4 - 8(2)^3 + 18(2)^2 = 24$ meters

$s(3) = (3)^4 - 8(3)^3 + 18(3)^2 = 27$ meters

$s(6) = (6)^4 - 8(6)^3 + 18(6)^2 = 216$ meters

(b.) Find the velocity at time t.

$s(t) = t^4 - 8t^3 + 18t^2$

$v(t) = s'(t) = 4t^3 - 24t^2 + 36t$

(c.) Find the velocity at $t = 2$ and $t = 4$.

$v(2) = s'(2) = 4(2)^3 - 24(2)^2 + 36(2) = 8$ meters/second

$v(4) = s'(4) = 4(4)^3 - 24(4)^2 + 36(4) = 16$ meters/second

(d.) When is the particle at rest?

The particle is at rest when $v(t) = s'(t) = 4t^3 - 24t^2 + 36t = 0$

$v(t) = s'(t) = 4t(t^2 - 6t + 9) = 0$

$v(t) = s'(t) = 4t(t-3)^2 = 0$

$t = 0$ and $t = 3$

The particle is at rest when $t = 0$ and $t = 3$.

A. (continued)

(e.) When is the particle moving in the positive direction? Negative direction?

The particle is moving in the positive direction when
$v(t) = s'(t) = 4t^3 - 24t^2 + 36t = 4t(t-3)^2 > 0$

This happens as long as $t > 0$.

(f.) Find the total distance traveled on the intervals $[0,1]$, $[0,2]$, $[0,3]$, and $[0,6]$.

$|s(1) - s(0)| = |11 - 0| = 11$ meters

$|s(2) - s(0)| = |24 - 0| = 24$ meters

$|s(3) - s(0)| = |27 - 0| = 27$ meters

$|s(6) - s(0)| = |216 - 0| = 216$ meters

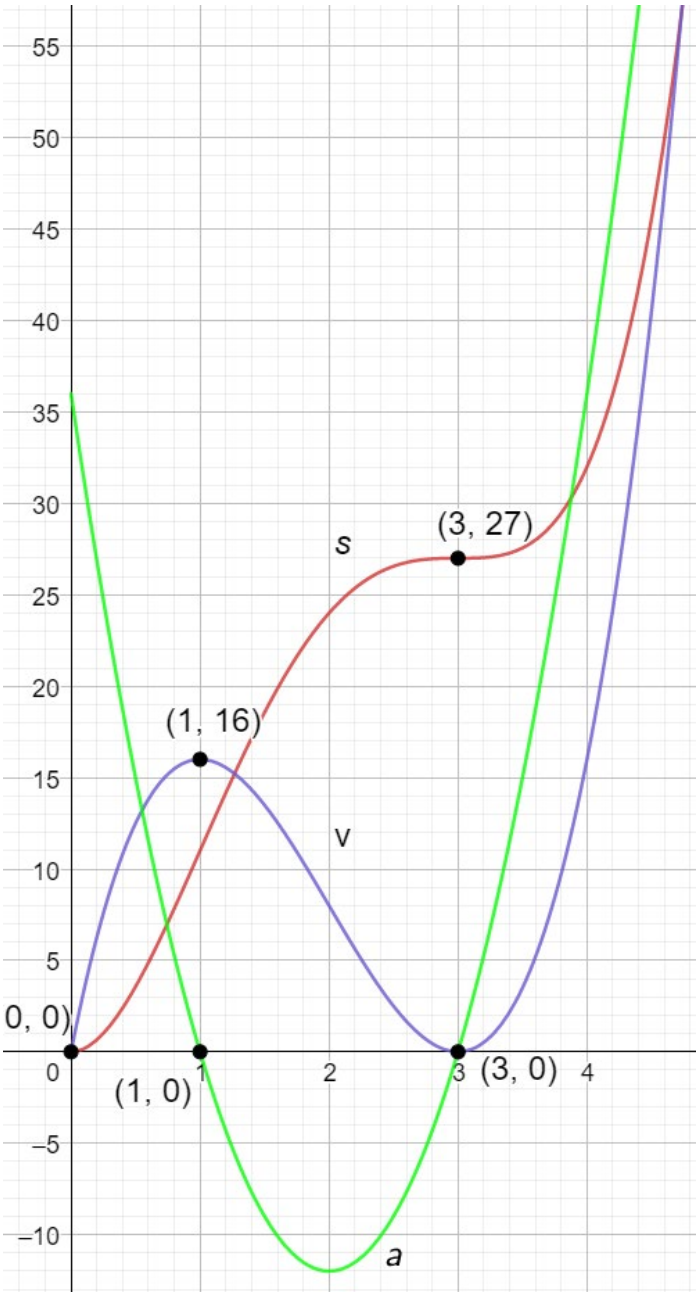
(g.) Draw a diagram to represent the motion of the particle. (Similar to Figure 2 on page 225)

(h.) Find the acceleration at time t.

$a(t) = v'(t) = s''(t) = 12t^2 - 48t + 36$

A. (continued)

(i.) Graph the position, velocity, and acceleration functions.



(j.) When is the particle speeding up? When is it slowing down?

The particle is speeding up when $0 < t < 1$ and $t > 3$.

The particle is slowing down when $1 < t < 3$.

Section 3.9 page 249 # 7

Section 3.11 page 265 # 39